

9.Synchronous action teaching

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1.Purpose

Using the LAN TCP remote communication function, one Muto can remotely control another Muto.

2.Communication principles

The principle of this experiment is to use one of the Muto as a server, open the TCP network service, receive data, and perform operations; while the other Muto serves as the client, that is, the operator, to read the angle value of the servo, and then sends the angle of the servo to the server through the TCP service; at this time, the client has been rapidly reading the angle value of the servo and sent it to the server. The server receives and immediately updates the angle of the servo, thereby realizing remote control of two hexapod robots. The function of synchronizing actions.

3.Preparation before experiment

1. This experiment involves two Muto robots with the same hardware. Except for the different running programs, everything else can be the same. If the Muto uses the system image that comes with the factory, please end the factory program that starts automatically at boot before performing this experiment.
2. Ensure that both Muto robots are connected to the same LAN, that is, connected to the same router (either wired or wireless), and can communicate within the LAN.
3. Since there are two Muto robots involved here, in order to easily distinguish them, we call the hexapod robot that is the server A hexapod robot (the controlled party), and the hexapod robot that is the controller is called A Muto robot. B Muto robot (free to move). Since the movement of Muto A depends on the movement of Muto B, please set both Muto robots to their default postures before starting the program.

4.Code

【Server code: A Muto robot】

Please see the content of the following path for the specific code:

```
~/muto/Samples/AI_Samples/12_sync_movement/A_robot_follow.py
```

【Client code: B Muto robot】

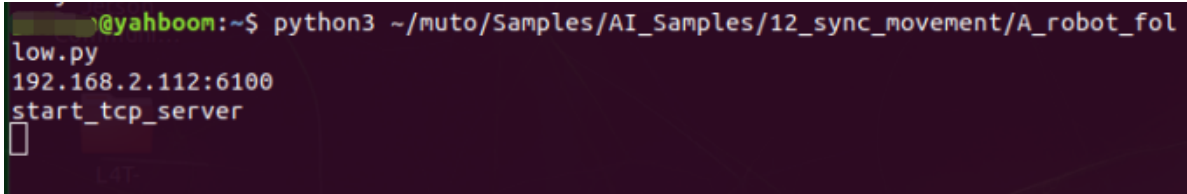
Please see the content of the following path for the specific code:

```
~/muto/Samples/AI_Samples/12_sync_movement/B_robot_ctrl.py
```

5.Operational experiments and experimental purposes

1. Open the terminal of the A Muto robot and run the A_robot_follow.py program. Then write down the printed IP address.

```
python3 ~/muto/Samples/AI_Samples/12_sync_movement/A_robot_follow.py
```



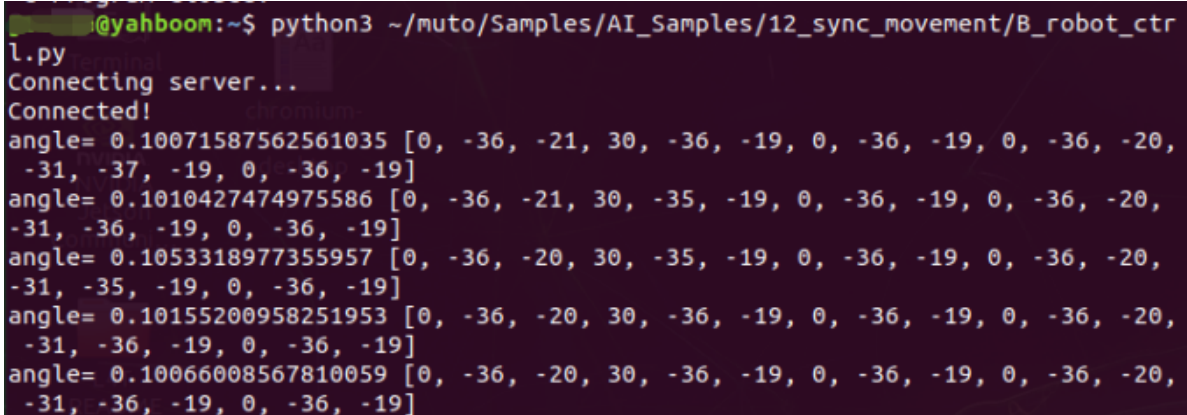
```
@yahboom:~$ python3 ~/muto/Samples/AI_Samples/12_sync_movement/A_robot_follow.py
192.168.2.112:6100
start_tcp_server
█
```

2. Open the terminal of B Muto robot and replace the IP address noted in the previous step to the ip address in B_robot_ctrl.py.

```
# 根据A_Robot的IP地址修改以下参数
# Modify the following parameters based on the IP address of A_Robot
ip = '192.168.2.112'
port = 6100
robot_init()
try:
    ready()
    connect_tcp_server(ip, port)
except:
    g_bot.load_leg(1)
    g_bot.zero_reset()
    waitClose(g_socket)
    del g_bot
    print(" Program closed! ")
```

3. After the modification is completed, save and exit, and then run the B_robot_ctrl.py program.

```
python3 ~/muto/Samples/AI_Samples/12_sync_movement/B_robot_ctrl.py
```



```
@yahboom:~$ python3 ~/muto/Samples/AI_Samples/12_sync_movement/B_robot_ctrl.py
Connecting server...
Connected!
angle= 0.10071587562561035 [0, -36, -21, 30, -36, -19, 0, -36, -19, 0, -36, -20, -31, -37, -19, 0, -36, -19]
angle= 0.1010427474975586 [0, -36, -21, 30, -35, -19, 0, -36, -19, 0, -36, -20, -31, -36, -19, 0, -36, -19]
angle= 0.1053318977355957 [0, -36, -20, 30, -35, -19, 0, -36, -19, 0, -36, -20, -31, -35, -19, 0, -36, -19]
angle= 0.10155200958251953 [0, -36, -20, 30, -36, -19, 0, -36, -19, 0, -36, -20, -31, -36, -19, 0, -36, -19]
angle= 0.10066008567810059 [0, -36, -20, 30, -36, -19, 0, -36, -19, 0, -36, -20, -31, -36, -19, 0, -36, -19]
```

Note that you must first run the `A_robot_follow.py` program of the A Muto robot, and then run the `B_robot_ctrl.py` program of the B Muto robot. The B Muto robot terminal will print the connection information and the sent data, and the communication between the two Muto robots will officially begin.

At this time, you can see that the state of the A Muto robot is the same as that of the B Muto robot, and you can move the leg1 leg of the B Muto robot by rotating it, and the leg1 leg of the A Muto robot will also move with it.

Through such TCP transmission commands, the effect of synchronized movements of two Muto robots can be achieved. When turning the servo angle of the Muto robot, please be careful that the angle range does not exceed the control range of the servo.